COMPUTER FUNDAMENTAL LAB REPORT # 2

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DEPARTMENT	**ELECTRICAL (Power)**
SECTION	**"A"**
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SUBJECT	**COMPUTER PROGRAMMING**

1. Objectives

The primary goal of this session was to establish a fundamental understanding of **C++** syntax Key learning outcomes included:

- Implementing the basic **boilerplate structure** of a C++ program.
- Managing **Standard I/O** (Input/Output) using cin and cout.
- Applying appropriate **Data Types** and **Arithmetic Operators**.
- Practicing industry-standard **indentation** and **documentation** (comments).

2. Technical Summary

A. Basic Syntax & I/O

Every functional C++ program requires the #include <iostream> header and a main() function.

- **cout <<**: Used for displaying data.
- **cin >>**: Used for capturing user input.
- **endl**: A manipulator used to insert a new line.

B. ***Data Types***

C++ requires explicit declaration of variables. The following were utilized:

- **int**: For whole numbers.
- **double**: For high-precision decimal values.
- **char**: For single characters (using single quotes).
- **bool**: For logical True/False.
- **string**: For text sequences.

C. ***Arithmetic Operators***

Calculations are performed using:

- Addition (+), Subtraction (-), Multiplication (*).
- **Division (/)**: Note that integer division truncates decimals.
- **Modulus (%)**: Returns the remainder of a division.

EXAMPLE # 1

```
int main() {
    //DIFFERENT TYPES K VARIABLE DICLARE KRNY K LIYE
    int a = 10;
    float floatVar = 5.5f;  //DECIMAL VALUES  K LIYE
    double doubleVar = 20.99; //LONG DECIMAL VALUE K LIYE
    char charVar = 'A'; //  CHARACTER LIKHNY K LIYR
    bool boolVar = true;

    // Printing values, sizes, and memory locations
    cout << "INT VARIABLE" << endl;
    cout << "Value: " << a << endl;
    cout << "Size: " << sizeof(a) << " bytes" << endl;
    cout << "Memory Address: " << &floatVar << endl << endl;

    cout << "FLOAT VARIABLE" << endl;
    cout << "Value: " << floatVar << endl;
    cout << "Size: " << sizeof(floatVar) << " bytes" << endl;
    cout << "Memory Address: " << &floatVar << endl << endl;

    cout << "DOUBLE VARIABLE" << endl;
    cout << "Value: " << doubleVar << endl;
    cout << "Size: " << sizeof(doubleVar) << " bytes" << endl;
    cout << "Memory Address: " << &doubleVar << endl << endl;

    cout << "CHAR VARIABLE" << endl;
    cout << "Value: " << charVar << endl;
    cout << "Size: " << sizeof(charVar) << " bytes" << endl;
    cout << "Memory Address: " << (void*)&charVar << endl << endl;

    cout << "BOOL VARIABLE" << endl;
    cout << "Value: " << boolVar << endl;
    cout << "Size: " << sizeof(boolVar) << " bytes" << endl;
    cout << "Memory Address: " << &boolVar << endl;

    return 0;
```

EXAMPLE # 2

```
#include <iostream>
using namespace std;
int main()
    {
        float pi; // FLAOT USE KIA HA DECIMAL VALUE K LIYE
        pi=3.14; // VALUE DECLARE
        cout<<"value of pi =\t"<<pi; // DISPLAY


return 0;
}
```

EXAMPLE # 3

```
#include <iostream>
using namespace std;
int main()
    {
        double F; // WORDS LIKHNY K LIYE
        float C; // DECIMAL VALUE K LIYE
        cout<<"enter temperature in celcius ";  // DISPLAY
        cin>>C;
    F = (C * 9/5) + 32;  // FORMULA CONVERRSION K LIYE
        cout<<F <<"  farheight";


return 0;
}
```

END